

Benchmarking Open-Source LLMs on CV Question Answering Pipelines Using a Fixed Human-Annotated Dataset

Master's Thesis

by Fahim Morshed

Supervised by **Dr. Rahat Ibn Rafiq**

Agenda

1. **Motivation & Problem Statement**
2. **Dataset Creation & Annotation**
3. **System Architecture & Pipeline Strategies**
4. **Experiment Design**
5. **Evaluation Framework**
6. **Results & Analysis** (RQ1–RQ4)
7. **Hardware Finding**
8. **Real-World Application: Admission Assistant**
9. **Limitations & Future Work**
10. **Conclusion**

Motivation

- Organizations process **hundreds to thousands** of CVs for hiring decisions
- Manual CV screening is **time-consuming, inconsistent, and subjective**
- Open-source LLMs are rapidly improving, but CV understanding still lacks a standardized benchmark centered on CV QA [1-5]
- Existing QA benchmarks such as **SQuAD, TriviaQA, DocVQA, and InfographicVQA** do not target the **structured + unstructured nature of CVs** [1-4]
- To our knowledge, no publicly available **human-annotated CV QA dataset** exists for systematic evaluation [1-5]

Gap: *How well can open-source LLMs extract and reason over information in real-world CVs — and which pipeline strategy works best?*

Research Questions

1. **RQ1:** How do retrieval strategies affect CV QA accuracy?
2. **RQ2:** How do open-source LLMs of varying sizes (1.5B, 7B, 13B) compare on CV QA tasks?
3. **RQ3:** What question types and error patterns emerge across pipelines?
4. **RQ4:** How well do confidence scores and rationales match actual correctness?

Background: LLMs for Information Extraction

- **Large Language Models** have shown strong capabilities in reading comprehension and information extraction
- This study focuses on three widely used open-weight model families: **Qwen2.5, Mistral, and Llama 2** [12-14]
- Key challenge: **context window management** — CVs vary in length and structure
- Two paradigms:
 - **Full-context:** Feed entire document to the model
 - **Retrieval-Augmented Generation (RAG):** Retrieve relevant sections first, then generate

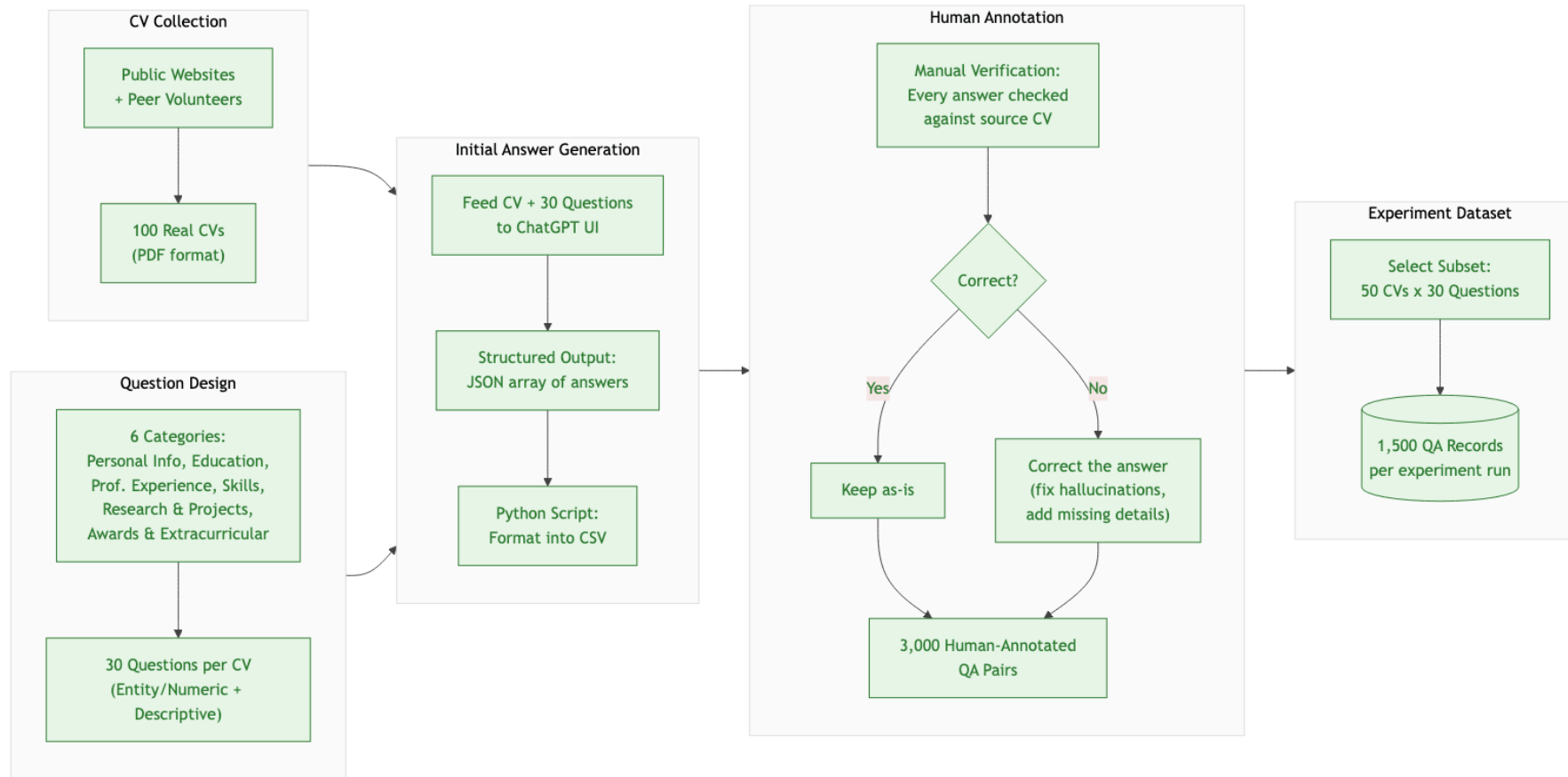
Related Work

Area	Key Work	Gap
Reading Comprehension	SQuAD, TriviaQA	General QA, not CVs
Document QA	DocVQA, InfographicsVQA	Visual docs, not CV QA
Resume Parsing	Resume-NER, rule-based NER	Extraction, not QA
RAG Systems	REALM, Lewis et al.	General retrieval, not CV QA
LLM Benchmarks	MMLU, HumanEval	General reasoning, not document QA

Representative references: [1] SQuAD, [2] TriviaQA, [3] DocVQA, [4] InfographicVQA, [5] Resume IE, [6] REALM, [7] RAG, [8] MMLU, [9] HumanEval, [15] RAG survey.

Our contribution: A **human-annotated** benchmark on **real CV PDFs** for end-to-end CV QA.

Dataset Creation Pipeline



Dataset Creation: CV Collection

- **Sources:**
 - Publicly available CVs from personal websites
 - Voluntarily shared by peers, classmates, and colleagues
- **100 CVs** collected in total
- All CVs in **PDF format** — real-world documents, not synthetic
- No standardized template — reflects real-world variability

Dataset Creation: Question Design

30 questions spanning 6 categories:

Category	Example Questions
Personal Information	Full name, contact details
Education	Institutions attended, GPA, degrees
Professional Experience	Current employer, years of experience
Skills	Programming languages, technical skills
Research & Publications	Publication count, research areas
Awards & Extracurricular	Awards, certifications

Answer Types: Entity/Numeric and Descriptive

Dataset Creation: Annotation Process

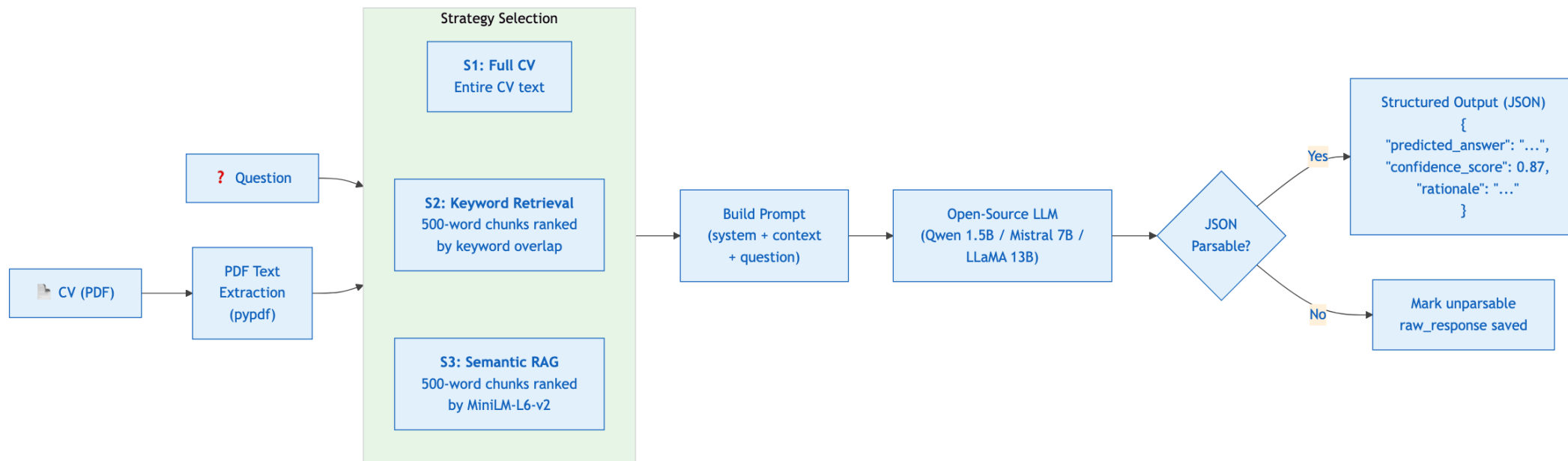
1. **Initial Generation:** Each CV + 30 questions fed to ChatGPT UI [10]
 - Output: structured JSON array of answers
 - Formatted into CSV using Python utility
2. **Human Verification & Correction:**
 - Every generated answer manually verified against the source CV
 - Corrections made where ChatGPT hallucinated or missed information
3. **Final Dataset:**
 - 100 CVs x 30 questions = **3,000 annotated QA pairs**
 - Experiment subset: **50 CVs x 30 questions = 1,500 records**

System Overview: Two-Phase Design

Our system is comprised of **two phases**:

1. **Generation Phase:** Open-source LLMs generate answers from CV content using one of three retrieval strategies
2. **Evaluation Phase:** Generated answers are evaluated using an LLM judge (Gemini 3 Flash) and validated through human spot-checking [11]

Phase 1: Generation Pipeline



Strategy 1: Full CV Context (S1)

- **Approach:** Entire CV text extracted via pypdf and injected into the prompt
- **Pros:** Maximum context — no information loss from retrieval
- **Cons:** Longer prompts, higher latency, may exceed context window for large CVs
- **Best for:** Questions requiring cross-section reasoning

The model sees everything in the CV — but has to find the relevant part itself.

Strategy 2: Keyword-Based Retrieval (S2)

- **Approach:** CV chunked into **500-word segments**; ranked by **keyword overlap** with question
- **Pros:** Reduced prompt size, focused context
- **Cons:** Keyword matching may miss semantically relevant sections
- **Retrieval:** Lexical word overlap scoring

Smaller context window — but relies on surface-level word matching to find relevant sections.

Strategy 3: Semantic RAG (S3)

- **Approach:** CV chunked into **500-word segments**; encoded with **all-MiniLM-L6-v2** embeddings
- **Retrieval:** Cosine similarity between question and chunk embeddings
- **Pros:** Captures meaning beyond keyword overlap
- **Cons:** Embedding quality dependency, additional compute
- **Caching:** Per-CV embedding cache for efficiency across questions

Semantic understanding for retrieval — but the embedding model is small and domain-general.

Models Under Study

Model	Parameters	Type	Key Characteristics
Qwen2.5-1.5B-Instruct	1.5B	Instruct	Smallest, fastest, resource-efficient
Mistral-7B-Instruct-v0.3	7B	Instruct	Mid-size, strong instruction following
LLaMA-2-13B-Chat-HF	13B	Chat	Largest, most capable, highest resource cost

- Represent a range of model sizes (1.5B, 7B, 13B)
- All open-source and locally deployable
- Cover different architecture families
- Enable analysis of **accuracy vs. efficiency tradeoffs**

Model refs: [12] Qwen2.5-1.5B-Instruct, [13] Mistral-7B-Instruct-v0.3, [14] Llama 2 / Llama-2-13B-Chat.

Experiment Design: Configurations

Strategy Comparison

Fixed model: Mistral-7B

Run	Strategy	Records
S1-Mistral	Full CV	1,500
S2-Mistral	Keyword	1,500
S3-Mistral	Semantic RAG	1,500

Model Comparison

Fixed strategy: S1 Full CV

Run	Model	Records
S1-Qwen	Qwen 1.5B	1,500
S1-Mistral	Mistral 7B	1,500
S1-LLaMA	LLaMA 13B	1,500

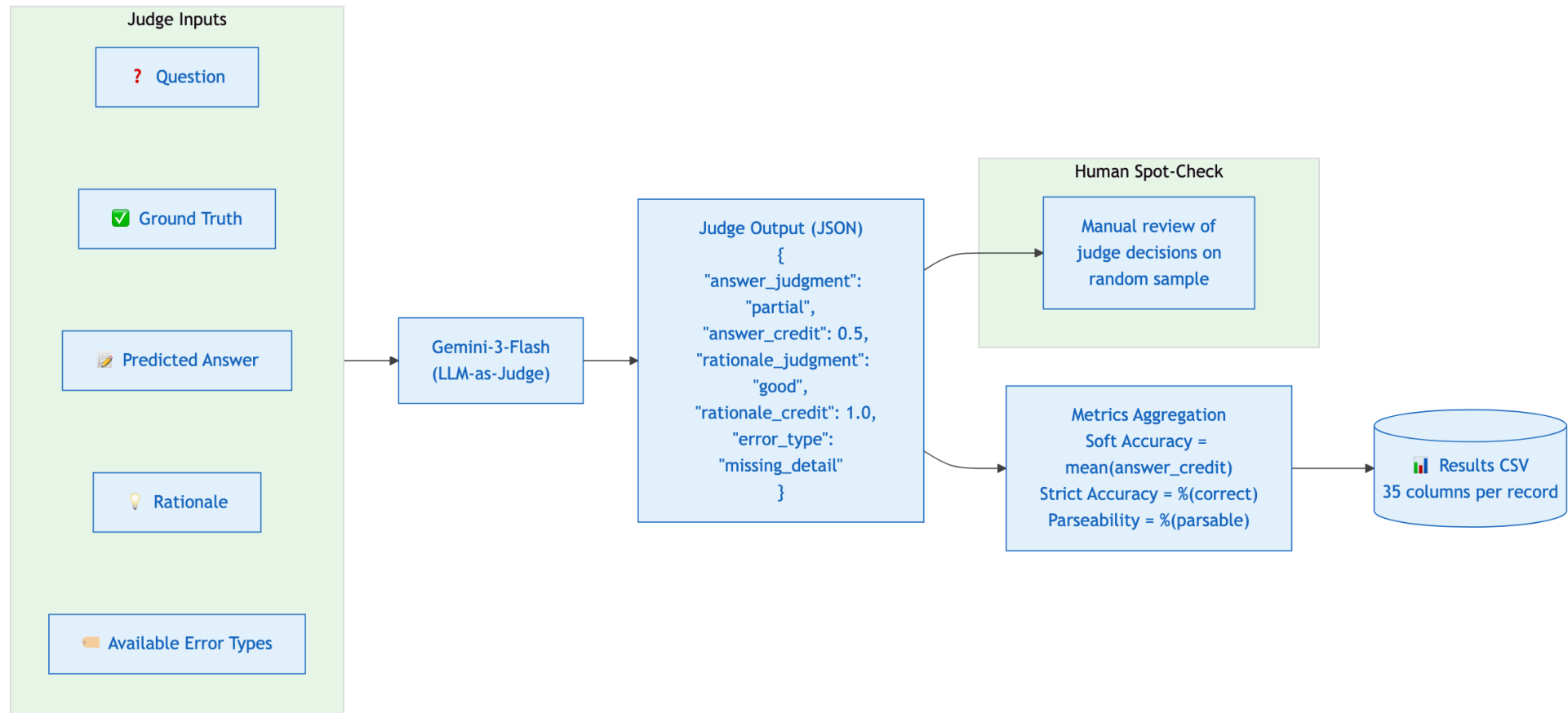
Hardware Comparison

Fixed strategy + model: S1 Mistral-7B

Run	Hardware	Records
S1-Mistral (CPU)	CPU	~1,500
S1-Mistral (GPU)	GPU	1,500

Total: 6 unique configurations x ~1,500 records = **~9,000 evaluated QA pairs**

Phase 2: Evaluation Pipeline



Evaluation: LLM-as-Judge

Judge Model: Gemini 3 Flash (Google) [11]

Input: question + ground_truth + predicted_answer + rationale

Output per record:

Field	Values
answer_judgment	correct / partial / incorrect
answer_credit	1.0 / 0.5 / 0.0
rationale_judgment	correct / partial / incorrect
rationale_credit	1.0 / 0.5 / 0.0
error_type	11 categories

Key Metrics: Soft Accuracy = mean(answer_credit), Strict Accuracy = %(correct), Parseability = %(valid JSON)

Evaluation: Error Taxonomy (11 Categories)

Error Type	Description
None (Correct)	No error — answer is correct
Not Answered	Model failed to provide an answer
Missing Detail	Correct direction but incomplete
Contradiction	Directly contradicts source
Wrong Entity	Extracted wrong entity from CV
Wrong Numeric	Incorrect numeric extraction
Wrong Boolean	Incorrect yes/no answer
Invented Value	Hallucinated for N/A questions
Extra Detail	Added information not in CV
Format Issue	Correct but formatted unusably
Other	Uncategorized errors

Results & Analysis

Master Results Table

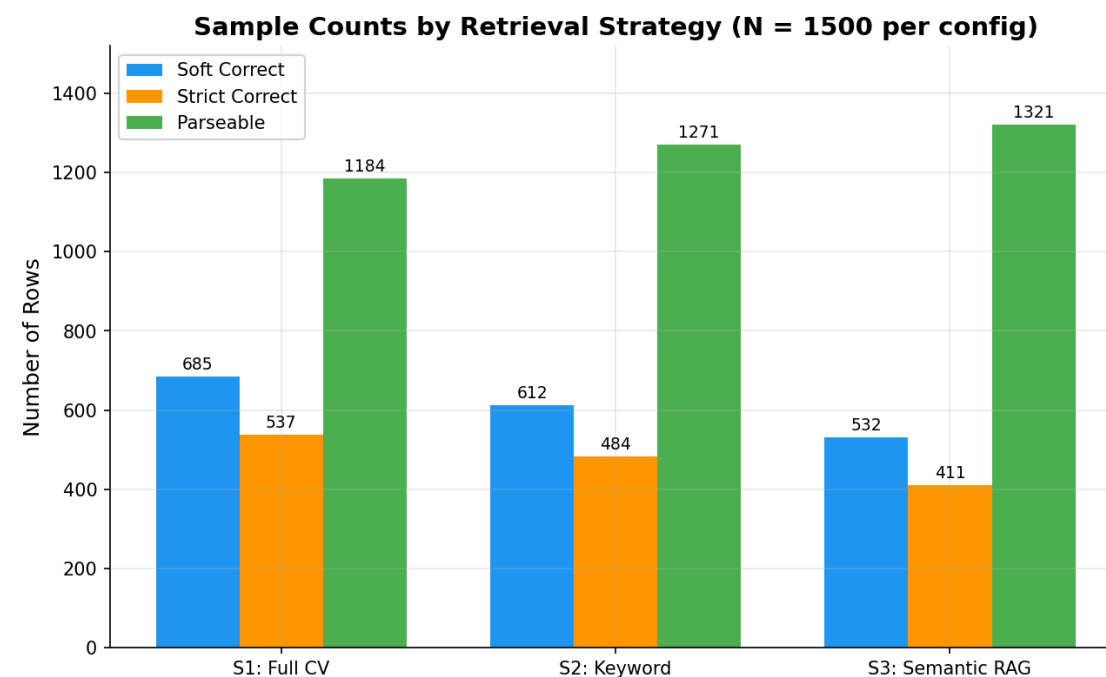
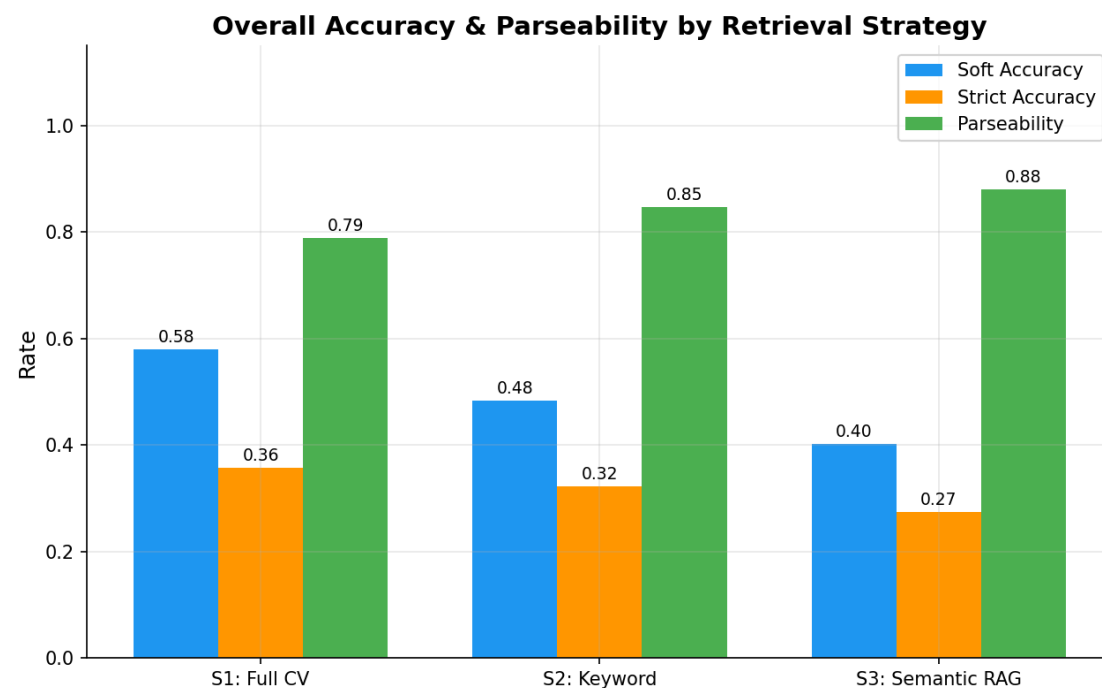
Configuration	Soft Acc.	Strict Acc.	Parseability	Avg Latency	Med. Latency	Avg In Tokens	Avg Out Tokens
S1 Qwen-2.5 (1.5B)	0.531	0.315	0.869	2.01s	2.02s	2,130	125
S1 Mistral (GPU)	0.579	0.358	0.789	2.12s	1.95s	2,525	86
S1 LLaMA-2 (13B)	0.551	0.315	0.816	5.44s	4.23s	2,593	119
S2 Mistral-7B	0.483	0.323	0.847	1.75s	1.43s	1,170	75
S3 Mistral-7B	0.403	0.274	0.881	1.94s	1.69s	990	72

Key finding: Mistral-7B with full CV context (S1) achieves the highest accuracy. Retrieval strategies (S2, S3) reduce latency but sacrifice accuracy. Bigger models (LLaMA 13B) don't outperform Mistral 7B.

RQ1: How Do Retrieval Strategies Affect Accuracy?

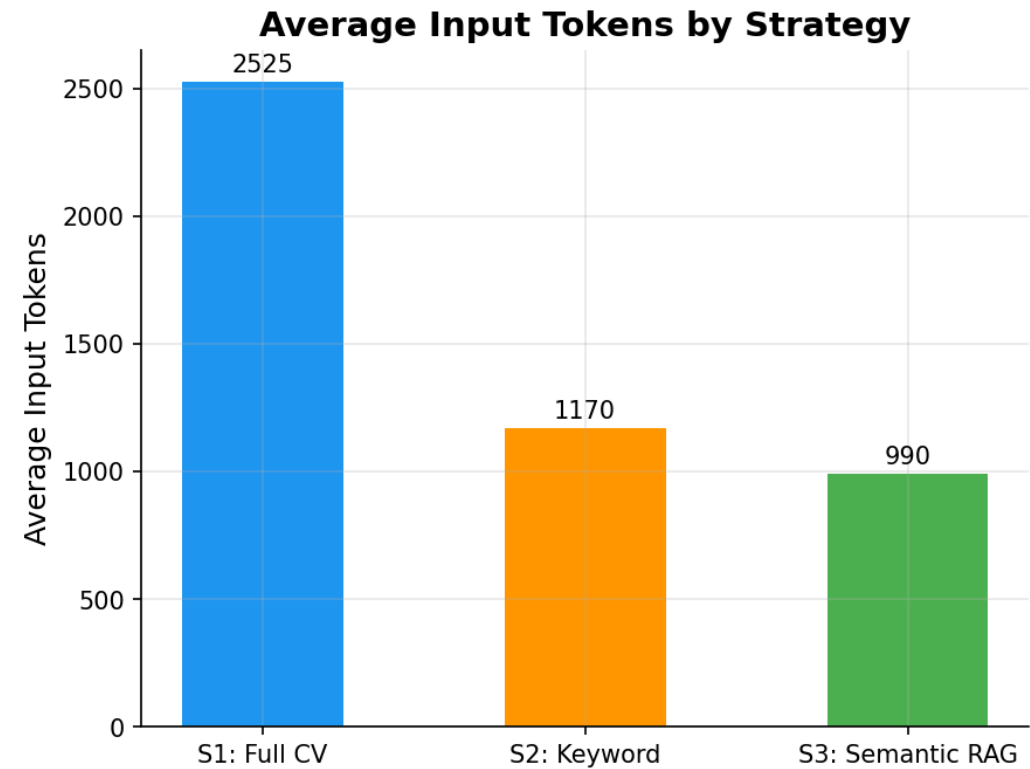
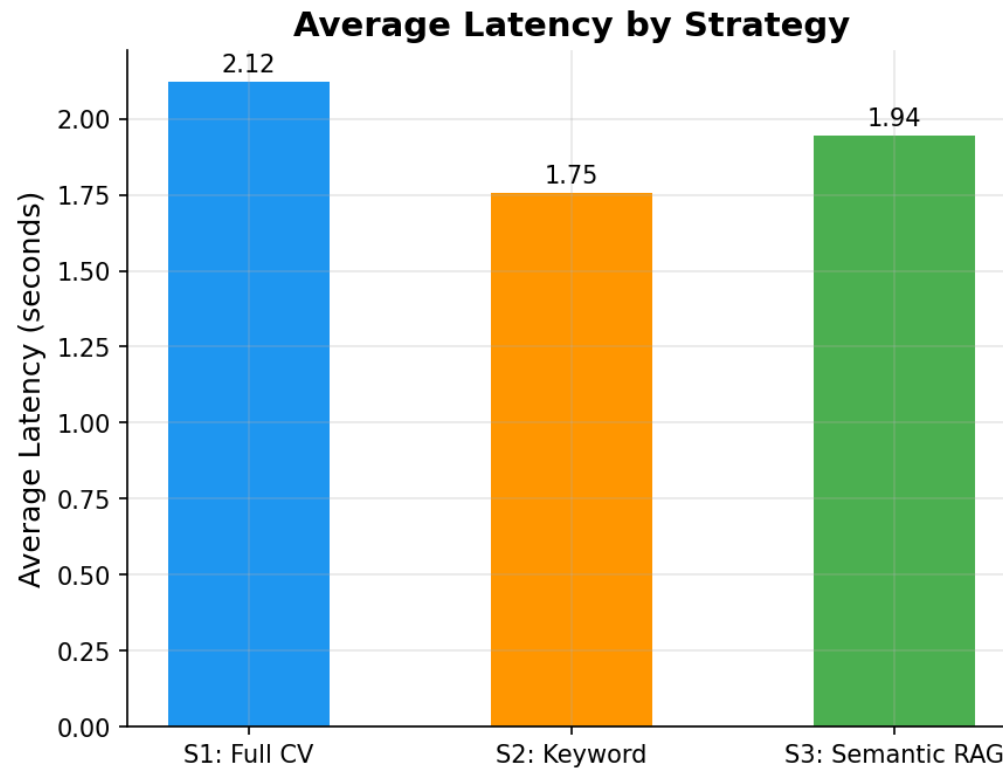
S1 vs S2 vs S3 — Model fixed to Mistral-7B

RQ1: Overall Accuracy & Sample Counts by Strategy



RQ1: Efficiency Comparison (Retrieval Strategies)

Efficiency Comparison by Retrieval Strategy



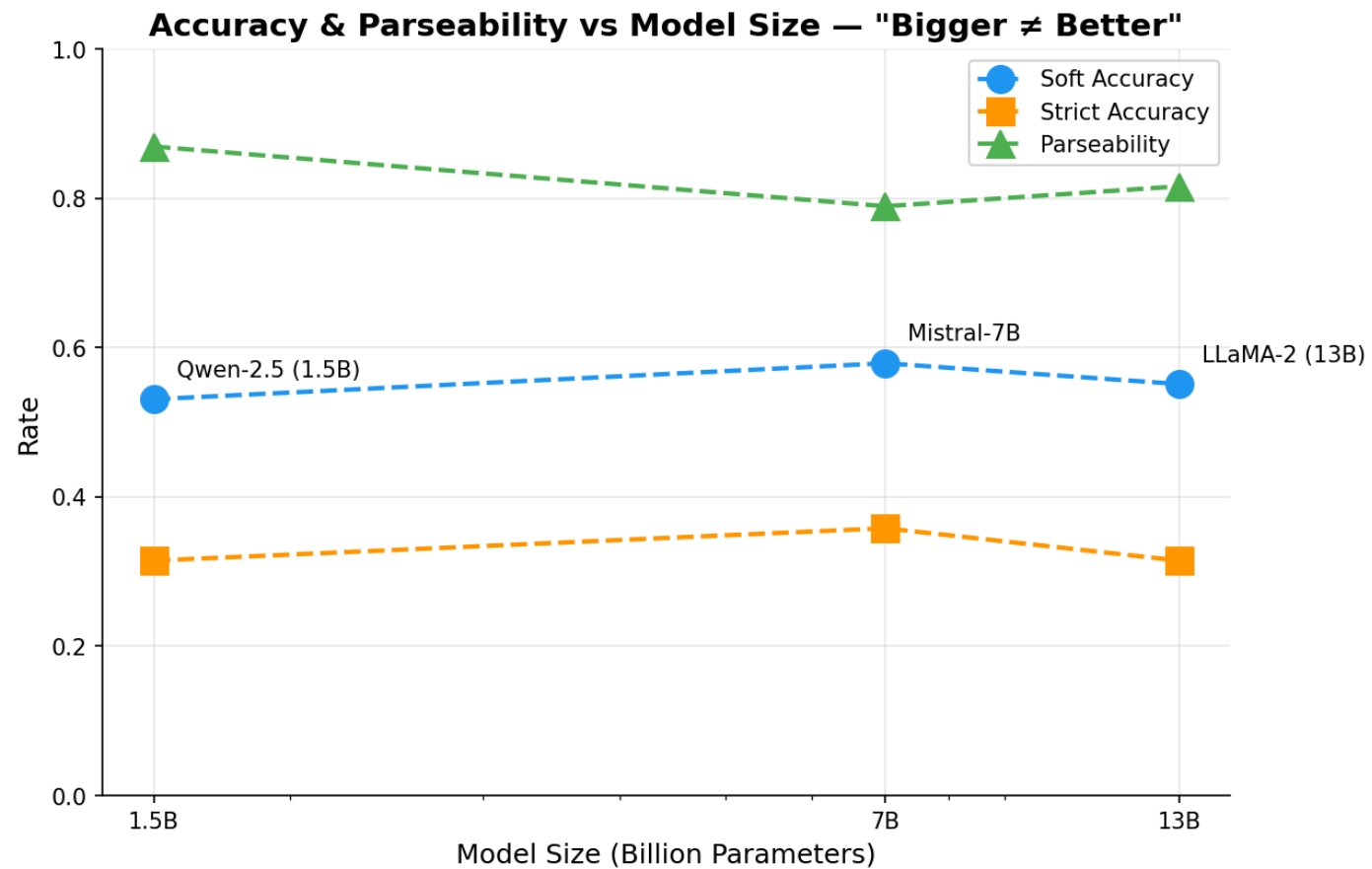
RQ1: Key Findings

- **S1 (Full CV) wins on accuracy:** 0.579 soft accuracy vs 0.483 (S2) and 0.403 (S3)
- **S3 (Semantic RAG) underperforms S2 (Keyword):** The small MiniLM-L6-v2 embedding model may not capture CV-domain semantics effectively
- **Retrieval saves tokens:** S2 uses ~1,170 input tokens vs S1's ~2,525 — a **54% reduction**
- **Trade-off:** Full context is worth the extra cost for CV QA accuracy

RQ2: How Do Models of Varying Sizes Compare?

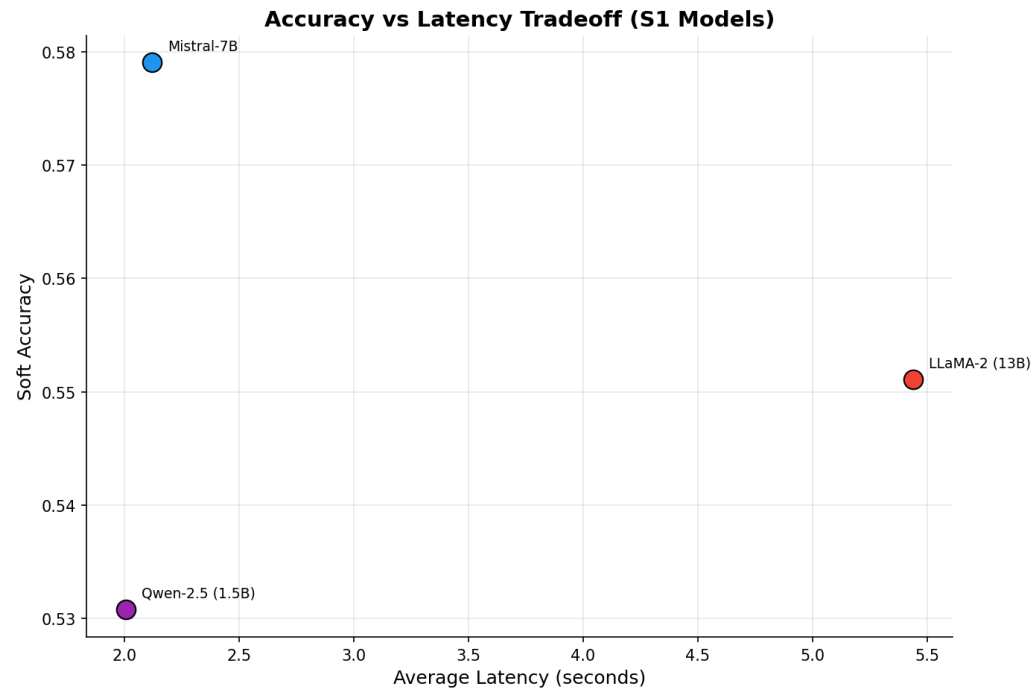
Qwen 1.5B vs Mistral 7B vs LLaMA 13B — Strategy fixed to S1

RQ2: Accuracy vs Model Size

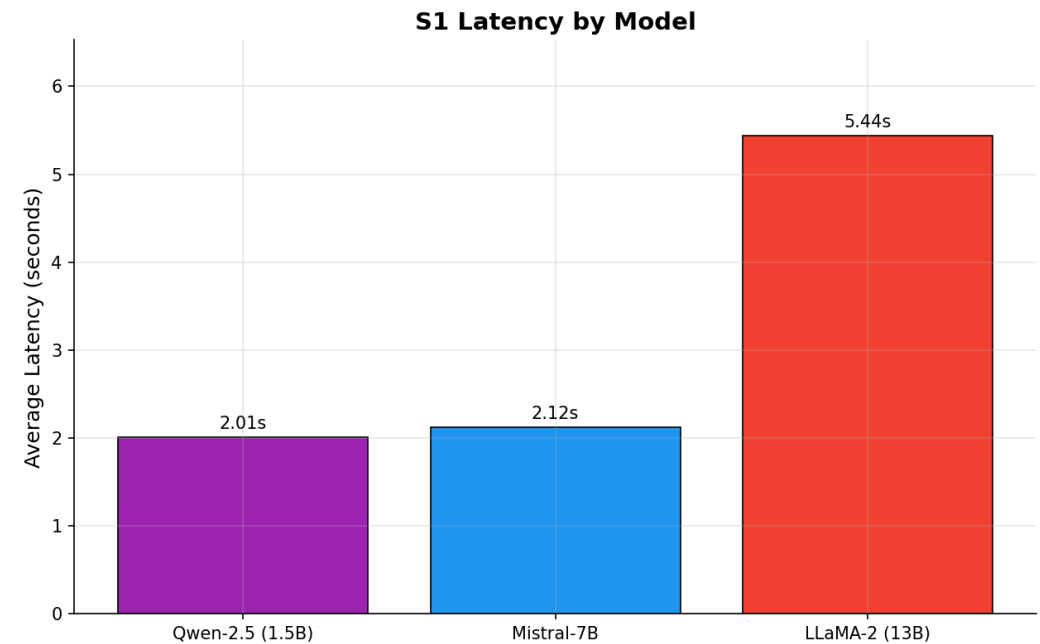


RQ2: Accuracy vs Latency Tradeoff

Accuracy vs Latency (S1 only)



Average Latency by Model (S1)



RQ2: Key Findings

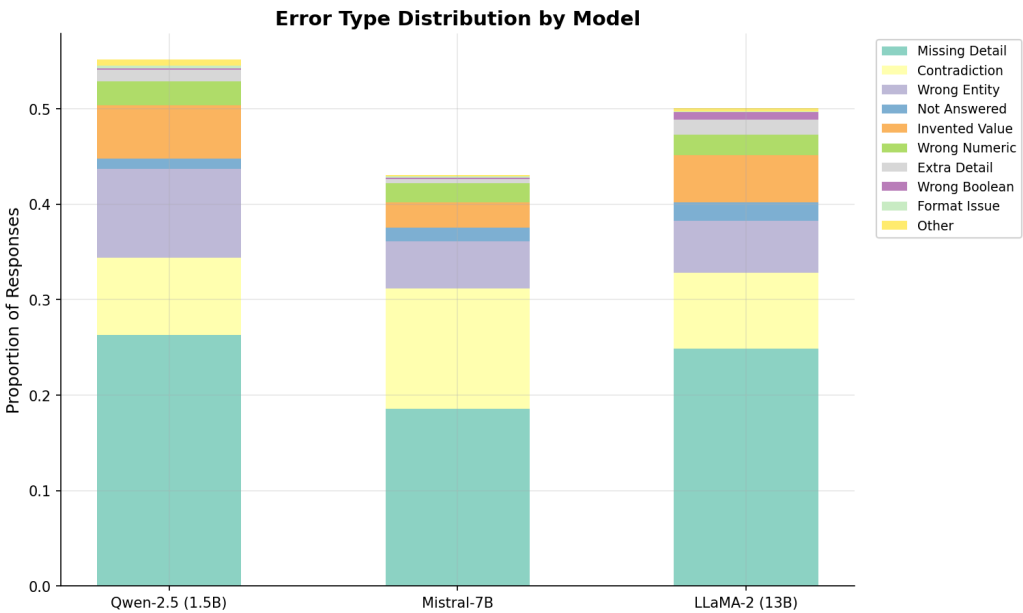
- **Bigger is NOT always better:** LLaMA-13B (0.551) does not outperform Mistral-7B (0.579) despite being nearly 2x the size
- **LLaMA-13B hit GPU memory limits** (99% utilization) — memory pressure likely caused degraded outputs
- **Qwen-1.5B is competitive:** 0.531 soft accuracy at 1/5th the size of Mistral, with comparable latency
- **Mistral-7B is the sweet spot:** Best accuracy, moderate latency (2.12s avg), balanced resource usage
- **Latency scales with size:** LLaMA averages 5.44s vs Mistral's 2.12s — a **2.6x slowdown**
- **Parseability varies:** Qwen (86.9%) > LLaMA (81.6%) > Mistral (78.9%)

RQ3: What Question Types and Error Patterns Emerge?

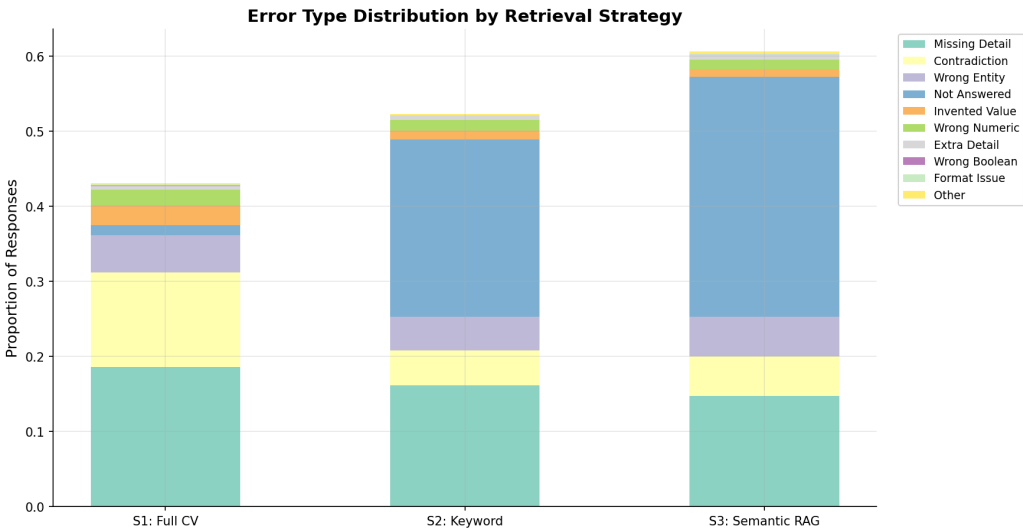
Cross-cutting analysis across all configurations

RQ3: Error Distribution by Model

By Model

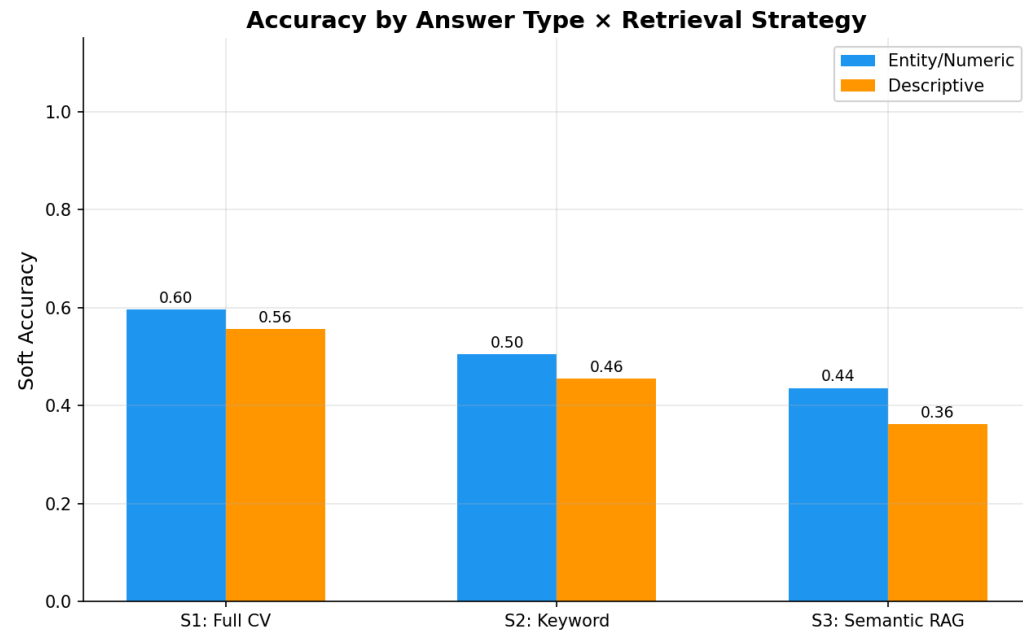


By Strategy

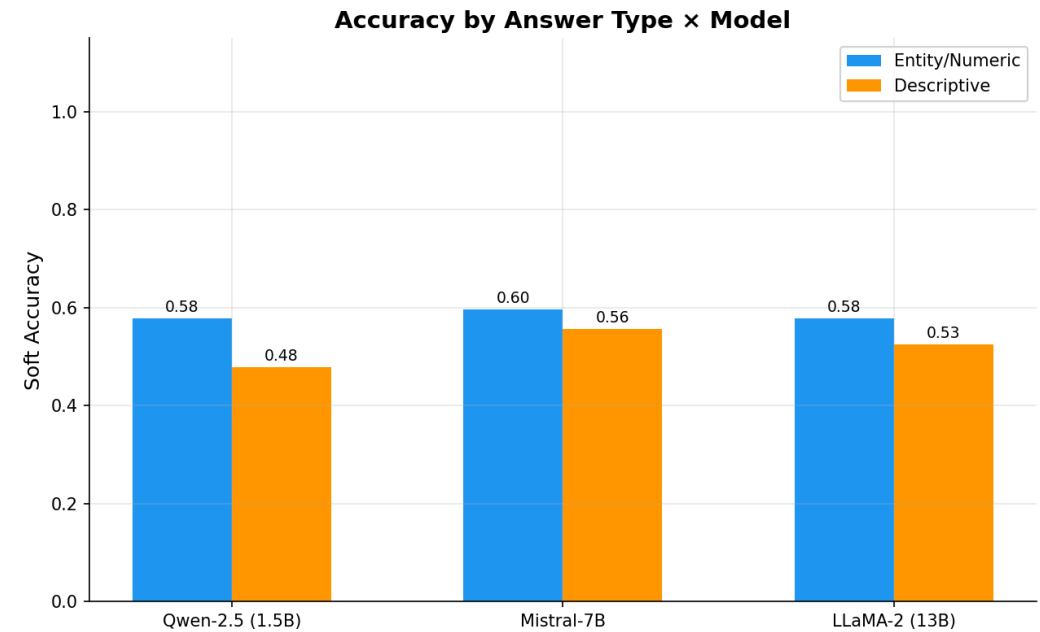


RQ3: Accuracy by Answer Type (Strategy x Model)

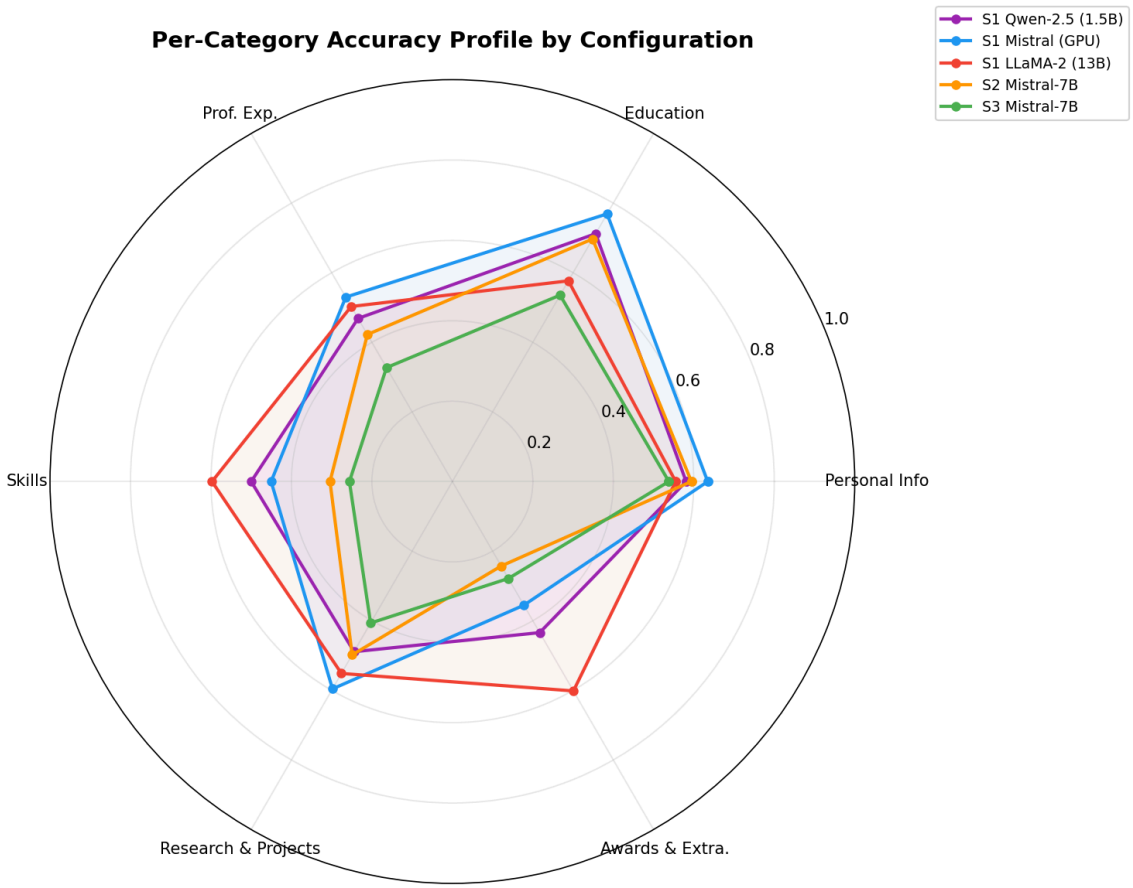
By Strategy



By Model



RQ3: Category Radar Chart



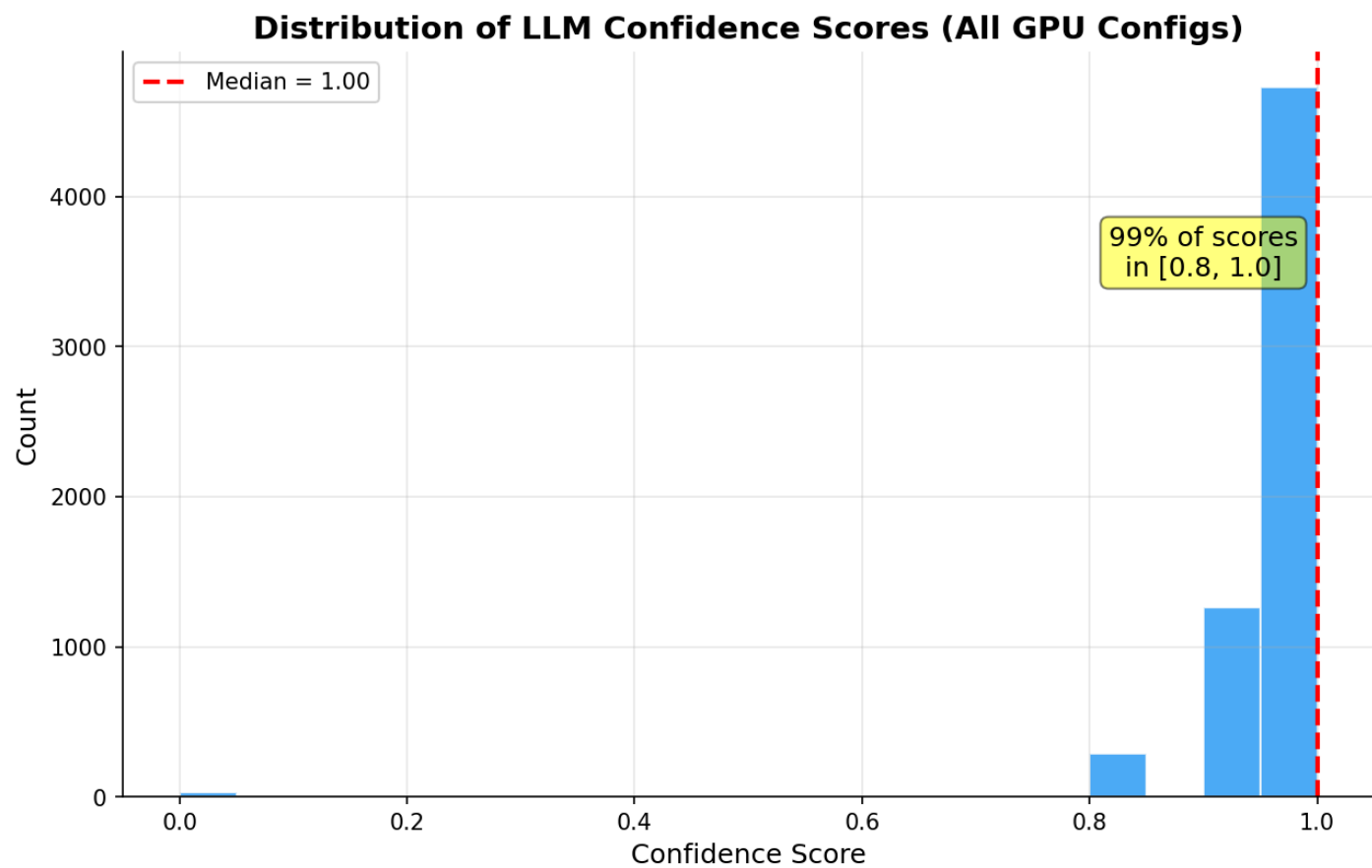
RQ3: Key Findings

- **"Missing Detail" is the dominant error** (1,511 total) — models often give partially correct answers
- **Entity/Numeric questions** are easier than Descriptive across all configs
- **Education** is the easiest category; **Awards & Extracurricular** is the hardest
- **LLaMA-13B performs best on Awards & Extracurricular** among compared configurations

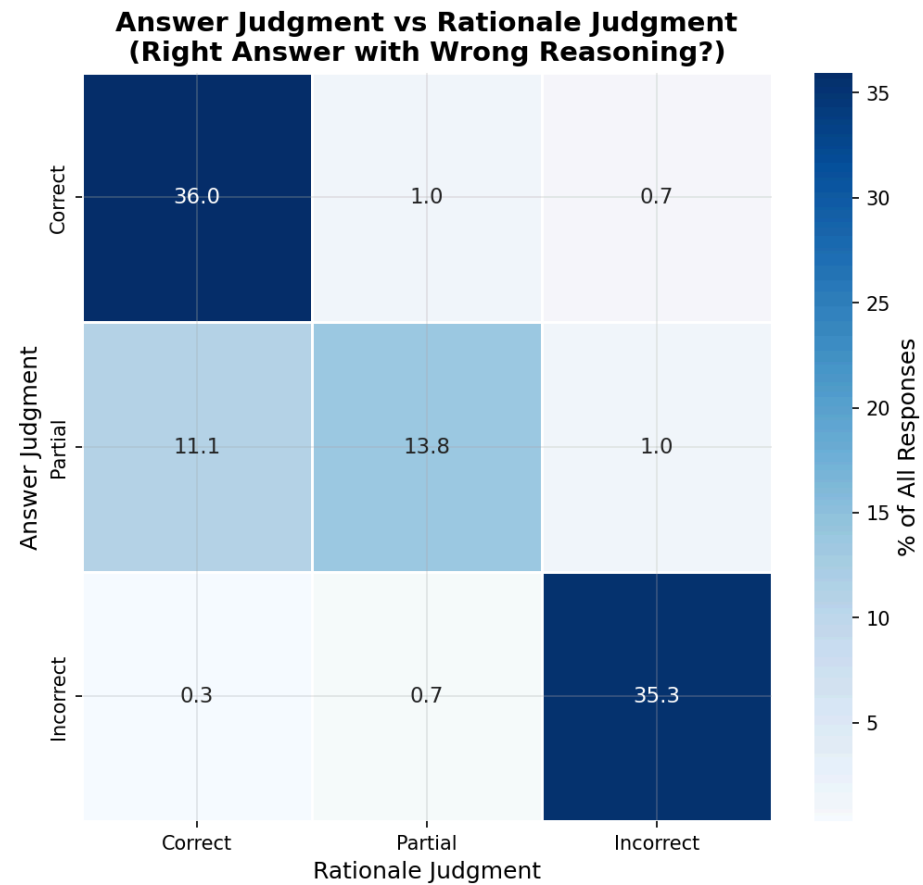
RQ4: Confidence Scores & Rationale Quality

Cross-cutting analysis across all configurations

RQ4: Confidence Score Distribution



RQ4: Answer vs Rationale Judgment

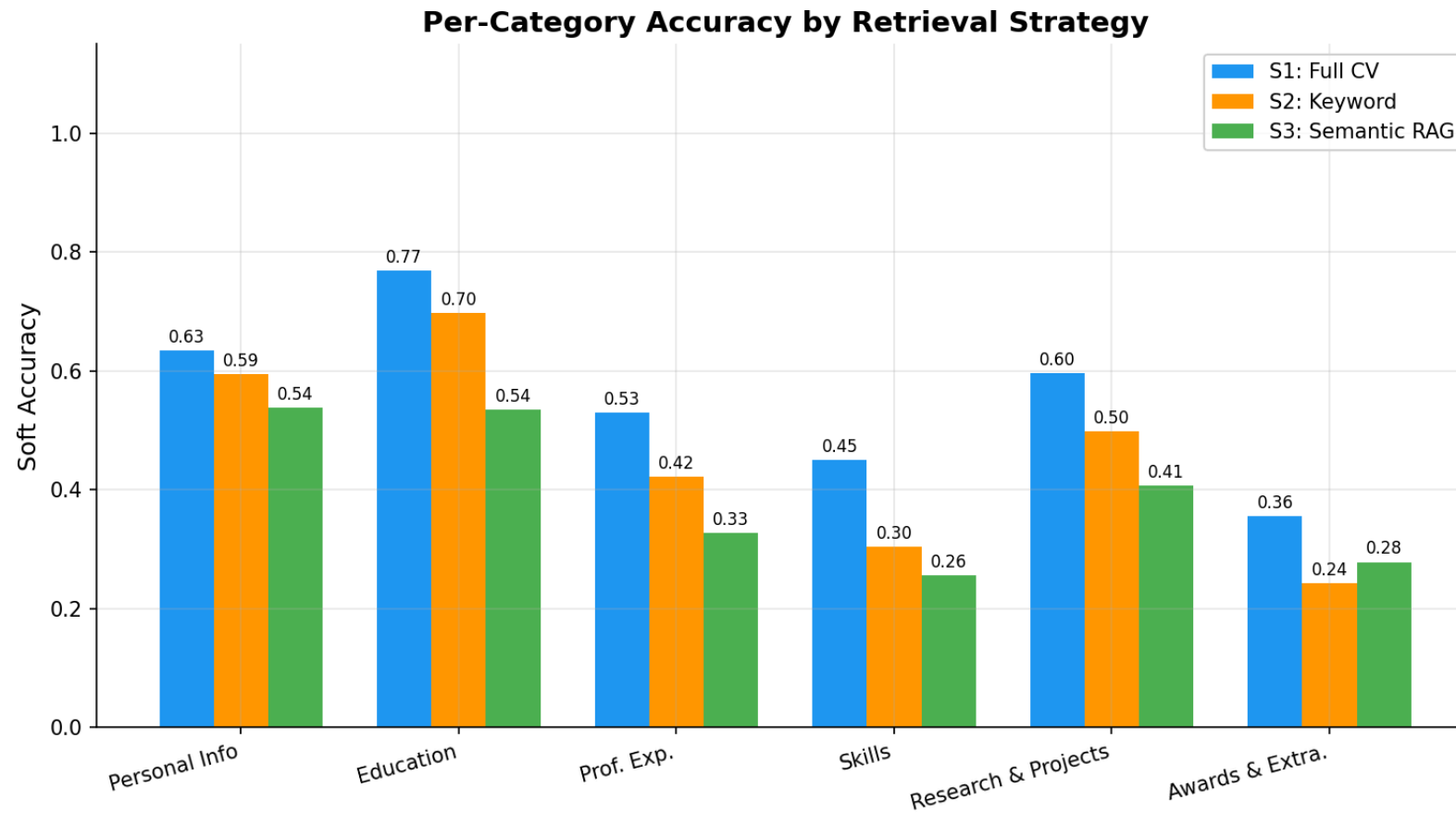


RQ4: Key Findings

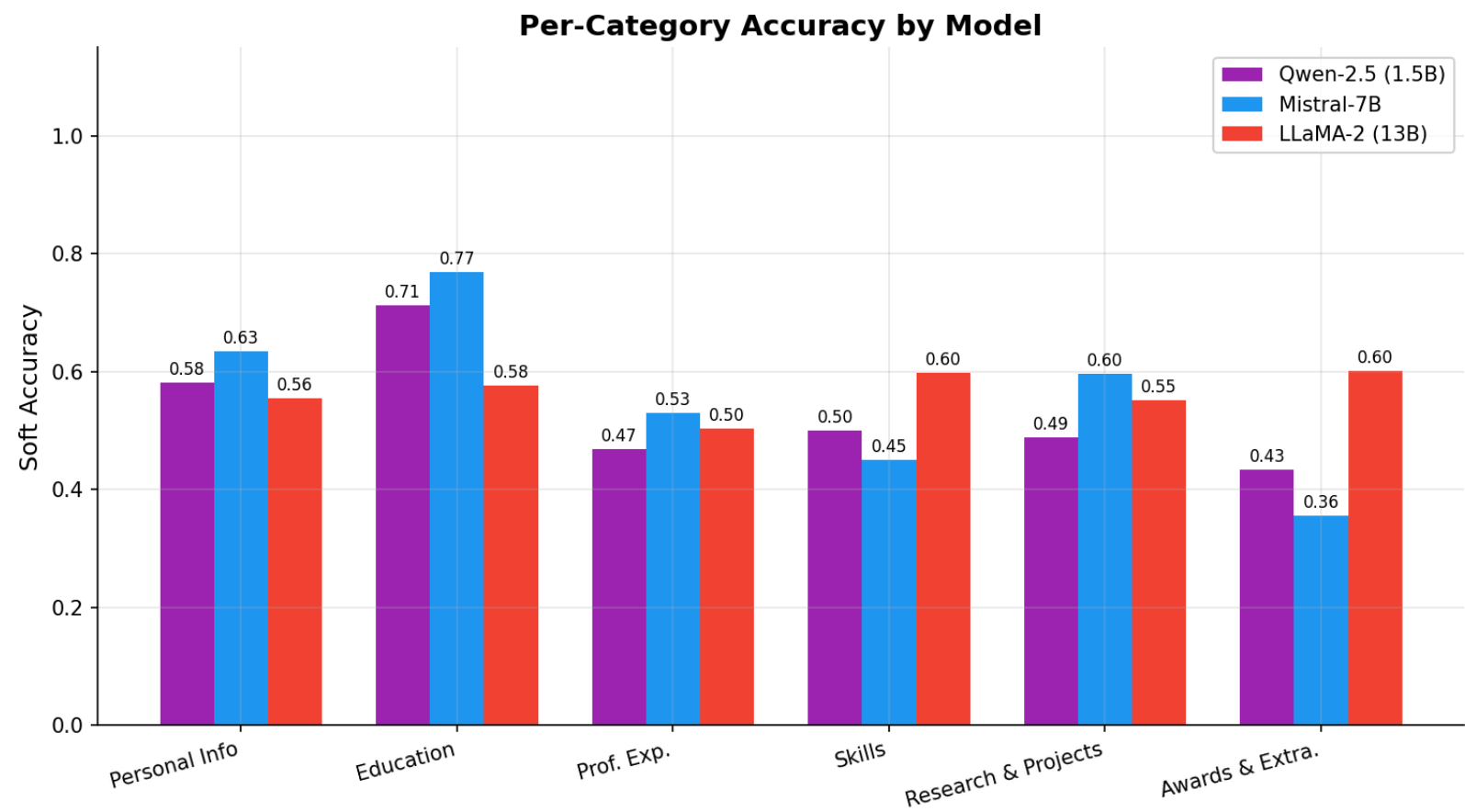
- **Models are overconfident:** Confidence scores cluster at 0.8–1.0 regardless of correctness
- **Poor calibration:** High confidence does NOT reliably indicate correct answers
- **Confidence is not a useful filter** for production use — cannot use it to flag uncertain answers
- **Rationale quality tracks answer quality:** Correct answers mostly have correct rationales
- **Right answer, wrong reason exists:** Some models get answers right through pattern matching rather than genuine understanding

Additional Findings

Per-Category Accuracy by Strategy

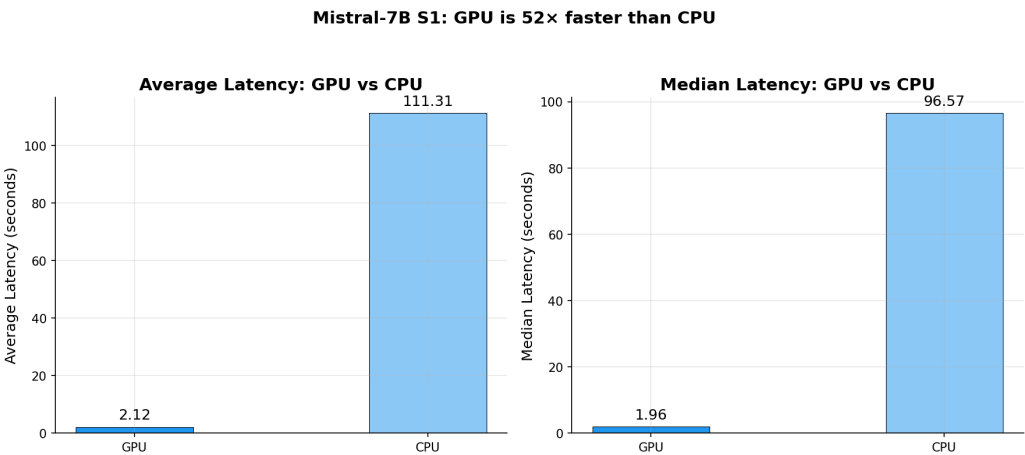


Per-Category Accuracy by Model

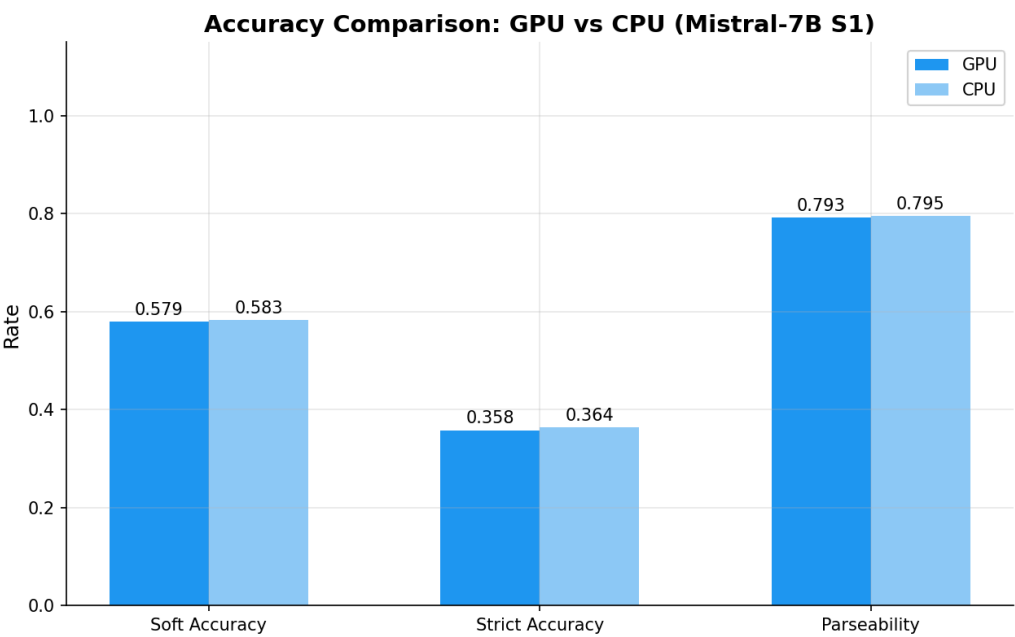


Hardware: Charts

Latency Comparison



Accuracy Comparison



Hardware: Key Findings

- **GPU is ~55x faster** than CPU for Mistral-7B inference
- **Accuracy is comparable** between GPU and CPU runs — the model produces similar quality outputs regardless of hardware
- **CPU is viable** for small-scale or offline processing where latency is acceptable
- **Implication:** Hardware choice affects throughput, not answer quality

Real-World Application: Admission Assistant

Admission Assistant: Overview

- Built as a **Graduate Assistantship project** at GVSU
- A web application to assist **Graduate Program Directors (GPDs)** in evaluating applicants
- Uses the **same strategy (S1: Full CV) and model (Mistral-7B-Instruct-v0.3)** that emerged as the best-performing in our experiments
- Demonstrates practical value of our benchmark findings

Key Features:

- Automated CV processing and QA extraction
- AI-suggested applicant ranking (leaderboard)
- Interactive CV question answering
- Role-based access control (GPD and Faculty roles)

Admission Assistant: Applicant Leaderboard

Admission Assistant

Leaderboard

Applicant Shares

Actions

Filter by Country...

Enable Multi-sorting on numeric columns

Status	Name	Institution	Country	GPA ↑↓	GRE ↑↓	TOEFL ↑↓	Major GPA ↑↓	DS & Algo ↑↓	Publicati
pending	Fahim Morshed	Bangladesh University of Engineering and Technology (BUET)	Bangladesh	3.47	0	-1	0	0	3
pending	Sumaiya Akter	Bangladesh University of Engineering and Technology (BUET)	Bangladesh	3.36	0	-1	0	0	2

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AI-suggested ranking based on extracted metrics: **GPA, GRE, TOEFL, Major GPA, DS & Algo scores** — all auto-extracted from CVs using the S1 full-context strategy.

Admission Assistant: Applicant Profile & QA

Admission Assistant
Leaderboard
Applicant Shares
Actions

Fahim Morshed
Email: f.morshed.077@gmail.com
Github: [fahshed](#)
LinkedIn: [Fahim Morshed](#)

3.47
CGPA

0
GRE

-1
TOEFL

Education

 **Bangladesh University of Engineering and Technology (BUET)**
Bachelor of Science in Computer Science and Engineering

Experience

 **IQVIA**
Software Developer (Current Position as of September 2022 - Present)

 **Department of CSE, BUET**
Software Developer (April 2022 - October 2022)

 **Sohopathi**
Junior Software Engineer (Part-time) (September 2020 - May 2021)

Curriculum Vitae

Fahim Morshed
Email: f.morshed.077@gmail.com | Mobile: +8801722947212 | LinkedIn: [Fahim Morshed](#) | Github: [fahshed](#)

What is the applicant's location?

The applicant's location is Dhaka, Bangladesh, as stated in their education and research experience details.

Logged in as: Test GPD
Logout

Applicant profile with **education**, **experience**, and an interactive QA panel — ask any question about the CV and get structured answers.

Admission Assistant: CV Viewer & QA

Admission Assistant

Leaderboard

Applicant Shares

Actions

Logged in as: Test GPD

Logout

• Replaced manual efforts in the exam evaluation pipeline by automated solutions.

• Wrote codes for React reusable components, Laravel eloquent queries and logics in API, MSSQL DB stored procedures.

• Presented progress to government stakeholders.

Junior Software Engineer (Part-time)

September 2020 - May 2021

Employer: [Sohopathi](#)

• Developed and maintained React-native mobile applications.

• Integrated APIs in react-native front-end and leveraged state management of redux-toolkit.

• Designed responsive web pages to properly show across multiple devices, e.g. mobile, tab, desktop.

• Extensively worked in every stages of MERN stack.

PROJECTS

Logistic Regression and Adaboost | *Machine Learning Project*

2022

• Implemented Logistic Regression and Adaboost from scratch.

Kemon Achen | *Software Development Project*

2022

• A mental healthcare mobile app using react native as front end and node as backend.

Attacks and AES | *Computer Security Projects*

2021

• Three fundamental Computer Security projects: ARP cache poisoning + Man in the middle attack, Cross site scripting (XSS) attack and Advanced Encryption Standard (AES) algorithm.

Rikshaw Fare Meter | *Micro-controller Project*

2019

• Built a fare calculating device with multiple sensors for a local vehicle called Rikshaw.

AWARDS

• Dean's List (Level-4): Bangladesh University of Engineering and Technology (BUET)

• Board Merit Scholarship: Government of Bangladesh (Year: 2012, 2014, 2016)

• Honourable Mention (3rd Position) at a Hackathon organized by BRAC University

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, PHP, SQL, C#, Kotlin, Bash, YAML.

Database: PostgreSQL, MongoDB, MS Sql Server

Frameworks: React, Angular, Express.js, ASP.NET, React-native, Native Android Tools: LASER, LaBSE, FAISS, Huggingface

REFERENCE

• [Dr. Anindya Iqbal](#)
Professor, Department of CSE, BUET
Email: anindya@cse.buet.ac.bd

• [Dr. A. B. M. Alim Al Islam](#)
Professor, Department of CSE, BUET
Email: alim_razi@cse.buet.ac.bd

• [Dr. Rifat Shahriyar](#)
Professor, Department of CSE, BUET
Email: rifat.shahriyar@gmail.com, rifat@cse.buet.ac.bd

The applicant's awards include: 1. Dean's List (Level-4) at Bangladesh University of Engineering and Technology (BUET) 2. Board Merit Scholarship from the Government of Bangladesh (received in the years 2012, 2014, 2016) 3. Honorable Mention (3rd Position) at a Hackathon organized by BRAC University

What frameworks does the applicant know?

The applicant, Fahim Morshed, is proficient in various frameworks such as React, Angular, Express.js, ASP.NET, React-native, as well as Native Android Tools like LASER, LaBSE, FAISS, and Huggingface. These frameworks are particularly relevant in web and mobile application development, machine learning, and natural language processing.

Is there any Professor among his references

Yes, there are three Professors listed as references in the information provided: Dr. Anindya Iqbal, Dr. A. B. M. Alim Al Islam, and Dr. Rifat Shahriyar.

Ask Another Question from Resume...

Ask!

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Side-by-side view: **rendered CV** alongside QA interactions. Faculty can ask follow-up questions directly from the applicant's resume.

49

Admission Assistant: CV Processing

Admission Assistant

Leaderboard

Applicant Shares

Actions

Process all unscanned CVs

List of Scanned CVs

Filename	Status
CV_Fahim_Morshed.pdf	Scanned

List of Unscanned CVs

Filename	Status
CV_Sumaiya_Akter.pdf	Scanned

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Batch processing pipeline — GPD uploads CVs and triggers automated scanning. Each CV is processed through the full-context QA pipeline.

Admission Assistant: Access Control & Sharing

Admission Assistant

Leaderboard

Applicant Shares

Actions

Applicant Shares

Shared by: Test GPD

Shared with: Test Faculty

Applicant: [Sumaiya Akter](#)

Comments:

Test GPD: Also this one!

Another com

Add Comment

Shared by: Test GPD

Shared with: Test Faculty

Applicant: [Fahim Morshed](#)

Comments:

Test GPD: Check out this applicant

Write your comment here

Add Comment

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Role-based access protocol: GPD processes and reviews CVs, then **shares specific applicant profiles** with faculty members. Faculty can view shared profiles and leave comments.

Discussion

Limitations

- **Hardware constraints:** LLaMA-13B pushed GPU to 99% utilization — results may improve on better hardware
- **Dataset scope:** 50 CVs in experiments — limited diversity in domains and formats
- **Model selection:** Three models from a rapidly evolving landscape
- **PDF extraction:** pypdf-based extraction loses formatting that visual models could leverage
- **Single-language:** English-only CVs and questions
- **Judge reliability:** Gemini-based evaluation may have biases (mitigated by spot-checking)
- **No task-specific fine-tuning:** Base models had instruction tuning, but none were fine-tuned on CV QA data
- **Retrieval chunking:** Fixed 500-word chunks may not align with CV section boundaries

Future Work

- **Expand dataset:** More CVs, more question types, multi-language support
- **Visual document understanding:** Layout-aware models (e.g., LayoutLM) that use formatting cues
- **Fine-tuning experiments:** Domain-specific fine-tuning on CV QA task
- **Newer models:** Evaluate emerging models (Mistral v0.4+, LLaMA 3, Qwen 2.5+, Phi-3)
- **Hybrid retrieval strategies:** Combine keyword and semantic retrieval
- **Human evaluation study:** Systematic inter-annotator agreement on judge quality
- **Production deployment:** Integrate findings into real-world decision support systems

Conclusion

- **Created** a novel human-annotated benchmark of **3,000 CV QA pairs** (100 CVs x 30 questions)
- **Designed** a modular evaluation framework with **3 pipeline strategies** and **3 model sizes**
- **Evaluated** ~9,000 records using **Gemini 3 Flash as LLM judge** across 6 configurations
- **Found** that **full CV context (S1)** consistently outperforms retrieval-heavy strategies for CV QA
- **Established Mistral-7B** as the best accuracy-efficiency tradeoff, while larger models did not yield better results
- **Identified** clear failure patterns: models miss detail, overstate confidence, and handle structured fields better than nuanced ones
- **Demonstrated** that hardware choice changes throughput by about **55x**, but not answer quality
- **Connected** the findings to a real-world admission assistant, showing the benchmark can inform practical system design

Acknowledgement

Citations

- [1] Rajpurkar et al. (2016). *SQuAD: 100,000+ Questions for Machine Comprehension of Text*. arXiv:1606.05250.
- [2] Joshi et al. (2017). *TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension*. arXiv:1705.03551.
- [3] Mathew, Karatzas, and Jawahar (2021). *DocVQA: A Dataset for VQA on Document Images*. arXiv:2007.00398.
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Thank You

Questions?